PART IIA, PAPER 3, SUPERVISION 3
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SOLUTIONS TO PROBLEMS

S3.1 Boys and girls: study hard and work harder, you may eventually get rich. Consider our old friend, the dataset `wage330.dta`. It provides data on the following characteristics for a sample of $n = 330$ young workers in the US in 1987:

`wage`	the individual's wage (in \$ per hour)
`male`	a 0-1 dummy variable, 1 for males
`school`	years of schooling
`exper`	years of work experience

(a) Generate a variable containing the logarithm of the hourly wage, name it `lwage`. Then compute a multiple regression explaining the logarithm of the wage as a function of gender, years of schooling and years of experience. Compare your estimates with those you computer in S2.6. Explain precisely what the coefficient on the schooling variable is measuring in the two regressions.

Solution: The regressions for wage and log wage are specified as follows:

$$\texttt{wage}_i = \beta_0 + \beta_1 \texttt{male}_i + \beta_2 \texttt{school}_i + \beta_3 \texttt{exper}_i + u_i, \tag{1}$$

$$\texttt{lwage}_i = \gamma_0 + \gamma_1 \texttt{male}_i + \gamma_2 \texttt{school}_i + \gamma_3 \texttt{exper}_i + v_i. \tag{2}$$

In the wage regression (1), the coefficients corresponding to continuous regressors measure the change in dollars in the hourly wage associated with a change in one unit of the corresponding regressor. In the log wage regression, the coefficients measure the percentage change (expressed as %/100) in the hourly wage associated with a one unit change in each regressor (semi-elasticity). Formally, for a continuous regressor x

$$\beta_x = \frac{\partial \texttt{wage}_i}{\partial x},$$
$$\gamma_x = \frac{\partial \ln(\texttt{wage}_i)}{\partial x} = \frac{1}{\texttt{wage}_i}\left(\frac{\partial \texttt{wage}_i}{\partial x}\right) = \left(\frac{\partial \texttt{wage}_i}{\texttt{wage}_i}\right)\left(\frac{1}{\partial x}\right).$$

Use the following `STATA` commands to estimate these regressions:

```
>> gen lwage = ln(wage)
>> reg wage male school exper
>> reg lwage male school exper
```

Table 1 then reports the two sets of estimates side by side in columns (1) and (2). Note that significance of individual variables as well as the joint regression significance are almost unaffected. The table was produced using the `STATA` command `esttab` for summarizing regression output in a publication style. The details of individual regressions are explained below.

```
>> gen lwage = ln(wage)
>> gen school2 = school^2
>> gen scmale = school * male
>> gen exmale = exper * male
```

```
>> eststo clear
>> eststo:  quietly reg wage male school exper
>> eststo:  quietly reg lwage male school exper
>> eststo:  quietly reg lwage male school school2 exper
>> eststo:  quietly reg lwage male school scmale exper
>> eststo:  quietly reg lwage male school scmale exper exmale
>> esttab, r2 ar2 scalars(F rss rmse)
```

Table 1: Output from wage and log wage regressions using `STATA`

	(1) `wage`	(2) `lwage`	(3) `lwage`	(4) `lwage`	(5) `lwage`
`male`	1.733^{***}	0.296^{***}	0.293^{***}	0.814^{**}	0.727
	(5.55)	(4.76)	(4.71)	(1.82)	(1.30)
`school`	0.768^{***}	0.143^{***}	-0.018	0.169^{***}	0.169^{***}
	(8.16)	(7.64)	(−0.14)	(5.79)	(5.76)
`school2`			0.007		
			(1.28)		
`scmale`				−0.044	−0.042
				(−1.17)	(−1.09)
`exper`	0.150^{*}	0.040^{**}	0.377^{**}	0.048^{**}	0.033
	(1.98)	(2.67)	(2.49)	(2.48)	(1.43)
`exmale`					0.008
					(0.26)
`const`	-5.221^{***}	−0.537	0.371	-0.832^{*}	−0.791
	(−3.67)	(−1.90)	(0.49)	(−2.19)	(−1.92)
N	330	330	330	330	330
R^2	0.209	0.181	0.187	0.186	0.186
adj. R^2	0.201	0.175	0.177	0.176	0.174
F-stat	28.67	24.33	18.69	18.61	14.86
SSR	2544.9	100.7	100.2	100.3	100.3
RMSE	2.784	0.556	0.555	0.555	0.556

t-statistics in parentheses

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

(b) By ordering the data (using the `sort` command), or otherwise, use the equation you have just estimated to test whether the effect of schooling on log wage is linear.

Solution: First, let's check whether we can spot any non-linearities using visual examination. Figure 1 presents a scatter plot of $\texttt{lwage}_i$ against $\texttt{school}_i$. Notice that school is not exactly a continuous variable, as can only take finite values ranging from 5 to 16 years of education. For each attainment level we see some variation in the values of log wages. The variance is increasing up until the level of education of 12 years (presumably high school level), and then starts falling as level of education rises from 13 to 16 years (university degrees). Conditional means withing each educational level are plotted as the red line. We can spot that from 6 to 12 years the means

fluctuate up and down sporadically, but then start increasing monotonically. There may be some non-linearity to the effect of schooling on log wages, but this needs to be tested formally.

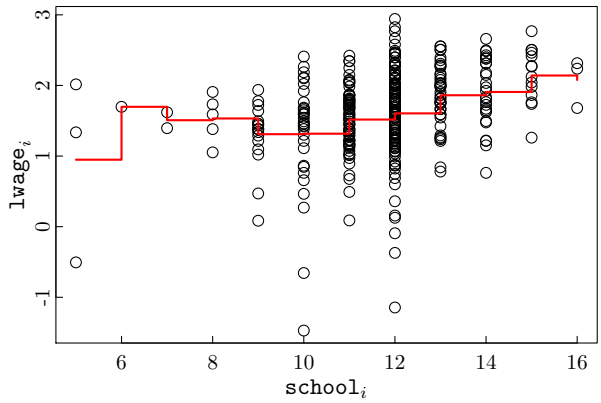

Figure 1: Scatter plot of $\texttt{lwage}_i$ against $\texttt{school}_i$.

We will use the Ramsey Regression Equation Specification Error Test (RESET), which is a general specification test for the linear regression model. The Ramsey RESET test tests whether non-linear combinations of the fitted values can help explain the response variable. The intuition behind the test is that if non-linear combinations of the explanatory variables have any power in explaining the response variable, then these variables will be significant. For a general multivariate regression setup with k explanatory variables

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + u_i, \quad i = 1, \ldots, n$$

consider the OLSE fitted values $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \cdots + \hat{\beta}_k x_{ki}$, where $\hat{\beta}_j$ are the OLS estimates. The Ramsey test then estimates the auxiliary regression with powers of the fitted dependent variable added to the original regression

$$y_i = \gamma_0 + \gamma_1 x_{1i} + \cdots + \gamma_k x_{ki} + \delta_1 \hat{y}_i^2 + \cdots + \delta_{k-1} \hat{y}_i^k + v_i. \tag{3}$$

An F-test is then conducted on the hypothesis $H_0 : \delta_1 = \cdots = \delta_{k-1} = 0$ with $k-1$ restriction, while the alternative is that at least of δ_j, $j = 1, \ldots, k-1$, is non-zero. Notice, for example, that the square of the fitted term implicitly includes all the squared regressors as well as their interactions, namely

$$\hat{y}_i^2 = \left(\hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \cdots + \hat{\beta}_k x_{ki}\right)^2 = \sum_{j=0}^{k} \beta_j^2 x_{ji}^2 + 2 \sum_{j>k} \beta_j \beta_k x_{ji} x_{ki}$$

This means that one single additional variable $\hat{y}_i^2$ takes into account the explanatory power of all possible combinations of the second-order polynomials in all the regressors, $x_{1i}, \ldots, x_{ki}$. This is much neater than addition all of these terms separately in the auxiliary regression and estimating an insanely large number of OLS coefficients. Now, recall that the null hypothesis is equivalent to no misspecification of the functional form, and the rejection of the null would suggest some non-linearity. Typically, `STATA` includes the second, the third and the fourth power of $\hat{y}_i$ and so tests for 3 restrictions $H_0 : \delta_1 = \delta_2 = \delta_3 = 0$ in the RESET test. This can be done done using

```
>> reg lwage male school exper
>> ovtest
out:  Ramsey RESET test using powers of the fitted values of lwage
out:  Ho:  model has no omitted variables
out:  F(3, 323) = 1.98
out:  Prob > F = 0.1167
```

This means that the test statistic $F = 1.98$ and the limiting distribution is $F_{3,323}$, while the p-value for the test is 0.1167, so we cannot reject the null at any significance level α lower than that. It is likely that the functional form of the regression is not misspecified. The Ramsey test suggests that the relationship is linear and that we don't need to add any powers of the explanatory variables in the regression.

(c) Construct an additional variable equal to `school`^{2}, and use this in a regression to test whether there are increasing or diminishing returns to schooling.

Solution: We want to estimate the following regression:

$$\texttt{lwage}_i = \beta_0 + \beta_1\texttt{male}_i + \beta_2\texttt{school}_i + \beta_3\texttt{school}_i^2 + \beta_4\texttt{exper}_i + v_i.$$

For this we can use the following `STATA` code:

```
>> gen school2 = school^2
>> reg lwage male school school2 exper
```

The estimation results are reported in column (3) of Table 1. Note that the the t-statistic for $H_0\colon \beta_3 = 0$ is very small and the null is not rejected, we conclude that the square of school is irrelevant and so the returns to schooling are neither increasing nor diminishing. An interesting thing to note that in addition to the square of school being insignificance, school also becomes insignificant.

(d) Now go back to the specification you estimated in (a). By constructing an interaction variable `scmale` = `school`×`male`, or otherwise, test whether the effect of additional schooling on log wage is the same for males and females.

Solution: We estimate the following regression:

$$\texttt{lwage}_i = \beta_0 + \beta_1\texttt{male}_i + \beta_2\texttt{school}_i + \beta_3\texttt{scmale}_i + \beta_4\texttt{exper}_i + v_i. \tag{4}$$

For this, use the following STATA code:

```
>> gen scmale = school * male
>> reg lwage male school scmale exper
```

The estimation results are reported in column (4) of Table 1. Note that the the t-statistic for $H_0 : \beta_3 = 0$ is very small and the null is not rejected. We conclude that returns to schooling are the same for males and females.

(e) By reordering the data, or otherwise, compute a Chow test statistic to test whether all the coefficients of an equation relating log wages to education and work experience are the same for males and females. [Hint: If you do this by estimating separate equations for males and females, i.e. by splitting the sample, explain carefully why you should not include the `male` variable in your sub-sample regressions. If you do this

using dummy variables, the `male` variable is already defined as a suitable intercept dummy, but you need to construct slope dummies.]

Solution: We estimate the following regression:

$$\texttt{lwage}_i = \beta_0 + \beta_1\texttt{male}_i + \beta_2\texttt{school}_i + \beta_3\texttt{scmale}_i + \beta_4\texttt{exper}_i + \beta_5\texttt{exmale}_i + v_i, \quad (5)$$

where $\texttt{exmale}_i = \texttt{exper}_i \times \texttt{male}_i$ is the interaction variable. We would like to test joint significance of the terms that involve $\texttt{male}_i$. Formally, we test $H_0\colon \beta_1 = \beta_3 = \beta_5 = 0$ against an alternative that at least one of them is not zero. We want to compute the F-test statistic by restricting the above regression to the regression that does not allow for gender-specific coefficients, namely.

$$\texttt{lwage}_i = \gamma_0 + \gamma_1\texttt{school}_i + \gamma_2\texttt{exper}_i + v_i. \quad (6)$$

Fortunately, `STATA` allows us to do the testing without explicit estimation of the restricted regression, instead we can use the following code:

```
>> gen exmale = exper * male
>> reg lwage male school scmale exper exmale
>> test male scmale exmale
out:  (1) exmale = 0
out:  (2) male = 0
out:  (3) scmale = 0
out:  F(3, 324) = 8.02
out:  Prob > F = 0.0000
```

The estimation results for the unrestricted regression are reported in column (5) of Table 1. Note that all three coefficients involving $\texttt{male}_i$ are insignificant. However, the F-test statistic for $H_0\colon \beta_1 = \beta_3 = \beta_5 = 0$ is 8.02 and the corresponding p-value is practically zero. Therefore, we can reject the null at any reasonable significance level. We conclude that the population coefficients of the combined returns to education and experience are not the same for males and females. Since in the previous part we concluded that returns to schooling alone do not differ significantly between genders, and the individual coefficients are insignificant, it is likely that the overall effect is driven by the interaction of the three effects: level effect, return to education and return to experience.

(f) By reordering the data, or otherwise, test whether the disturbances of the specification you estimated in section (a) are homoskedastic. What would you do if you found evidence of heteroskedastic disturbances?

Solution: We can perform a Breusch-Pagan test for heteroskedasticity (the null hypothesis of this test is that the disturbances are homoskedastic). This test makes a strong assumption about normality of errors. Be careful when implementing the test, since if this assumption is violated in the dataset, then the results of the Breusch-Pagan inference are invalidated. We will conduct two versions of the test, one that does not assume anything regarding the shape or origin of the eventual heteroskedasticity, and another one assuming that the $\mathbb{E}\left[u_i^2|\texttt{school}_i\right] = f\left(\texttt{school}_i\right)$, i.e. the variance of the error terms changes across the levels of schooling. The general setup for the Breusch-Pagan test is based on a auxiliary regression for the squares of the fitted residuals. First, the coefficients $\hat{\beta}_0$, $\hat{\beta}_1$, $\hat{\beta}_2$ and $\hat{\beta}_3$ in a linear model are estimated

using OLS, and then the fitted values of $\widehat{\texttt{lwage}}_i$ and fitted residuals $\hat{u}_i$ are computed:

$$\widehat{\texttt{lwage}}_i = \hat{\beta}_0 - \hat{\beta}_1\texttt{male}_i - \hat{\beta}_2\texttt{school}_i - \hat{\beta}_3\texttt{exper}_i.$$
$$\hat{u}_i = \texttt{lwage}_i - \widehat{\texttt{lwage}}_i.$$

Note that $\hat{u}_i^2$ is a measure of the error variance since the errors have mean zero. An auxiliary regression for $\hat{u}_i^2$ can the be formulated with either the fitted value $\widehat{\texttt{lwage}}_i$ as the explanatory variable, or the original regressors, depending on the hypothesized functional form of heteroskedasticity. We will consider the following possibilities:

$$\hat{u}_i^2 = \gamma_0 + \gamma_1\widehat{\texttt{lwage}}_i + \varepsilon_i \tag{7}$$
$$\hat{u}_i^2 = \gamma_0 + \gamma_1\texttt{school}_i + \varepsilon_i \tag{8}$$
$$\hat{u}_i^2 = \gamma_0 + \gamma_1\texttt{male}_i + \gamma_2\texttt{school}_i + \gamma_3\texttt{exper}_i + \varepsilon_i \tag{9}$$

Regression (7) suggests that $\mathbb{E}\left[u_i^2\right] = f\left(\widehat{\texttt{lwage}}_i\right)$ and we want to test $H_0\colon \gamma_1 = 0$, which implies homoskedasticity of errors, against $H_1 : \gamma_1 \neq 0$, which implies heteroskedasticity of errors. A similar interpretation can be given to regression (8). Finally, regression (9) suggests that $\hat{u}_i^2 = f\left(\texttt{male}_i, \texttt{school}_i, \texttt{exper}_i\right)$ and so we want to test joint significance $H_0\colon \gamma_1 = \gamma_2 = \gamma_3 = 0$ against the alternative that at least on coefficient is non-zero, implying heteroskedasticity. This can be done is STATA using the following code:

```
>> reg lwage male school exper
>> hettest
out:  Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
out:  Ho:  Constant variance
out:  Variables :  fitted values of lwage
out:  chi2 (1) = 7.65
out:  Prob > chi2 = 0.0057

>> hettest school
out:  Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
out:  Ho:  Constant variance
out:  Variables :  school
out:  chi2 (1) = 5.03
out:  Prob > chi2 = 0.0249
```

In both cases the hypothesis of homoskedasticity is rejected, but not so strongly when looking at schooling, so the heteroskedasticity in the model may be more strongly related to other variables. Or the Breusch-Pagan test with non-normality assumptions on the disturbances. Test specification (9) using the following code:

```
>> hettest male school exper, iid
out:  Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
out:  Ho:  Constant variance
out:  Variables :  male school exper
out:  chi2 (3) = 4.46
out:  Prob > chi2 = 0.2155
```

The default Breusch-Pagan test specified by `hettest` is a test for linear forms of heteroskedasticity. In this default form, the test does not work well for non-linear forms of heteroskedasticity. If heteroskedasticity is detected, there are several possible courses of action:

(1) Use White's heteroskedasticity test, as it relaxes assumption of normally distributed errors and allows for general non-linearity of heteroskedasticity.

(2) Respecify regression using a different functional form for the dependent variable. (but here it's already in logs).

(3) Scale the data by dividing all variables by `school`, since we have evidence that the variance of the unobserved is a function of schooling.

(4) Use heteroskedasticity-robust standard errors (HAC errors). In contrast to (1) and (2), this doesn't attempt to remove heteroskedasticity. Errors remain heteroskedastic, but the standard errors for the parameter estimates are estimated taking this into account, so that valid inference can be conducted. This is less efficient than modeling the heteroskedasticity, but more robust, as it doesn't need additional assumptions.

S3.2 Does your IQ let your parents' schooling interfere with your education? Download the dataset `wage2.dta`. Use the `STATA` commands `des` and `summ` to understand the structure of the data and the meaning of the variable labels. Now answer the following questions relating to performance on the IQ test for this sample of working-age women. (The IQ test was taken as an adult after the woman had completed her formal education.)

(a) Run a regression of IQ test score on parents education. What do you conclude?

Solution: Column (1) in Table 2 reports OLS estimates of the following regression

$$\texttt{IQ} = \beta_0 + \beta_1 \texttt{feduc}_i + \beta_2 \texttt{meduc}_i + u_i, \quad i = 1, \ldots, n. \tag{10}$$

The coefficients for father's and mother's education are positive and close to unity. Both are individually significant at 0.1%, as suggested by the p-values (three stars). Moreover, the regression coefficients are jointly significant, since the F-test statistic for the overall regression significance is very high, $F = 61.72$. The goodness-of-fit is relatively low at $R^2 = 0.147$.

The output table was produced using the `STATA` command `esttab` for summarizing regression output in a publication-style table. You can read about how to install and use this function in this online tutorial. The details of individual regressions are explained in the solutions below.

```
>> eststo:  quietly reg IQ feduc meduc
>> eststo:  quietly reg IQ feduc meduc educ
>> eststo:  quietly reg IQ educ if ((feduc != .)  & (meduc != .))
>> eststo:  quietly reg IQ feduc meduc educ if black == 0
>> eststo:  quietly reg IQ feduc meduc educ if black == 1
>> eststo:  quietly reg IQ feduc meduc educ feducb meducb educb
>> esttab, r2 ar2 scalars(F rss rmse)
```

Table 2: `STATA` output for regressions of `IQ` on parental and own education.

	(1) IQ	(2) IQ	(3) IQ	(4) IQ	(5) IQ	(6) IQ
feduc	1.02***	0.36*		0.18	0.70	0.19
	(5.40)	(2.06)		(1.07)	(1.09)	(1.04)
meduc	1.06***	0.61**		0.40*	0.75	0.42*
	(4.82)	(3.05)		(1.98)	(1.09)	(2.05)
educ		3.03***	3.55***	3.00***	2.48*	3.06***
		(13.22)	(17.08)	(13.53)	(2.26)	(13.80)
feducb						0.55
						(0.97)
meducb						0.33
						(0.55)
educb						−1.57***
						(−3.62)
const	80.21***	50.38***	53.66***	56.06***	42.45**	54.99***
	(38.45)	(17.19)	(18.67)	(19.14)	(3.41)	(19.12)
Sample	full	full	full	white	black	full
N	722	722	722	657	65	722
R^2	0.147	0.314	0.288	0.300	0.225	0.372
adj. R^2	0.144	0.311	0.287	0.297	0.187	0.367
F-stat	61.72	109.4	291.82	93.29	5.903	70.68
SSR	134273	107986	111952	86164	12368	98756
RMSE	13.67	12.26	12.47	11.49	14.24	11.75

t statistics in parentheses

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

(b) Suppose we want to know whether the only way parents education increases their daughters test score is through the daughters own education. Test that hypothesis using an F-test. What do you conclude from your test? Is there any policy recommendation from your results?

Solution: In order to test whether parental education has no effect on child's IQ beyond its influence of the child's education, we can estimate a regression of IQ on own education as well as father's and mother's education and test the joint significance of the latter two variables. They way to do that is to estimate an unrestricted regression with all three explanatory variables as well as the restricted model with only own education, i.e. where the coefficients of `feduc` and `meduc` are restricted to be zero:

$$\text{Unrestricted: } \texttt{IQ} = \beta_0 + \beta_1 \texttt{feduc}_i + \beta_2 \texttt{meduc}_i + \beta_3 \texttt{educ}_i + u_i, \quad RSS_u,\ R_u^2 \tag{11}$$

$$\text{Restricted: } \texttt{IQ} = \delta_0 + \qquad\qquad + \delta_3 \texttt{educ}_i + v_i, \quad RSS_r,\ R_r^2 \tag{12}$$

The estimation output for the unrestricted regression is presented in column (2), while the output for the restricted regression is in column (3). Note that since there are quite a few missing observations for `feduc` and `meduc`, we need to restrict the sample

for the restricted regression to only include those individuals where no observations on parental education are missing. For this we need to run STATA code

```
>> reg IQ feduc meduc educ
>> reg IQ educ if ((feduc!=.)  & (meduc!=.))
```

Formal testing involves constructing an F-test statistic for the following hypothesis, $H\colon\ (\beta_1 = 0) \cap (\beta_2 = 0)$ against $H_1\colon\ (\beta_1 \neq 0) \cup (\beta_2 \neq 0)$. For this we will need the sum of squared residuals SSR or R^2 from the restricted and the unrestricted models. The following F-test statistic has the $F_{q,n-k-1}$ distribution:

$$F = \frac{(SSR_r - SSR_u)/q}{SSR_u/(n-k-1)} = \frac{\left(R_u^2 - R_r^2\right)/q}{(1-R_u^2)/(n-k-1)} \sim F_{q,n-k-1}, \tag{13}$$

where $SSR_r = 111952.3$ and $R_r^2 = 0.288$ are the sum of squared residuals and R^2 for the restricted model in (3), while $SSR_u = 107986.9$ and $R_u^2 = 0.314$ are the sum of squared residuals and R^2 for the unrestricted model in (2), respectively, $q = 2$ is the number of restrictions, $k = 3$ is the number of explanatory variables in the unrestricted model, and $n = 722$ is the sample size. For the general case in (13), the rejection rule is $F > F_{1-\alpha,q,n-k-1}$, where $F_{1-\alpha,q,n-k-1}$ is the critical value of the $F_{q,n-k-1}$ distribution at the $1-\alpha$ probability. Pick $\alpha = 0.001$ and so our critical value is $F_{0.999,2,718} = 6.974$. The observed F-statistic is given by

$$F = \frac{\left(SSR^{(3)} - SSR^{(2)}\right)/q}{SSR^{(2)}/(n-k-1)} = \frac{(111952.3 - 107986.9)/2}{107986.9/718} = 13.18,$$

which is clearly greater than the critical value and so we reject the null at $\alpha = 0.001$. The same test can be done in STATA by doing an F-test for joint significance of `feduc` and `meduc` in regression (11) using the following commands:

```
>> reg IQ feduc meduc educ
>> test feduc meduc
```

This yield an F-test statistic of 13.18, same as our direct computations. We can conclude from these results that the father's and mother's education affects IQ of the child beyond their influence on the child's own education. Perhaps there is a direct genetic effect on intelligence. Notice from column (2) that mother's education has twice the partial effect of the father's education, and its significance is much higher. We can therefore suggest that the emphasis of the policy should be on raising educational level of women. For example, one could consider grants of scholarships women in the lower socioeconomic stratum.

(c) Regress IQ on father's education, mother's education and own education separately for two subsets of individuals, those with `black` $= 0$ and those with `black` $= 1$. Test the hypothesis that the coefficients in the two regressions are the same.

Solution: To conduct this test, in addition to the full-sample regression in column (2) we need to estimate two regressions, one for the sub-sample of non-black individuals and one for the sub-sample of black individuals, namely

$$\texttt{black} = 0 : \texttt{IQ} = \beta_0^w + \beta_1^w \texttt{feduc}_i + \beta_2^w \texttt{meduc}_i + \beta_3^w \texttt{educ}_i + u_i, \quad RSS_1 \tag{14}$$

$$\texttt{black} = 1 : \texttt{IQ} = \beta_0^b + \beta_1^b \texttt{feduc}_i + \beta_2^b \texttt{meduc}_i + \beta_3^b \texttt{educ}_i + u_i, \quad RSS_2 \tag{15}$$

The estimation can done in STATA using the following code:

```
>> reg IQ feduc meduc educ if black == 0
>> reg IQ feduc meduc educ if black == 1
```

The estimation outputs for these regressions are reported in columns (4) and (5) of Table 2, respectively. The corresponding SSR and R^2 are also provided. We want to test the the hypothesis of coefficient equality:

$$H_0\colon \left(\beta_0^w = \beta_0^b\right) \cap \left(\beta_1^w = \beta_1^b\right) \cap \left(\beta_2^w = \beta_2^b\right) \cap \left(\beta_3^w = \beta_3^b\right)$$
$$H_1\colon \left(\beta_0^w \neq \beta_0^b\right) \cup \left(\beta_1^w \neq \beta_1^b\right) \cup \left(\beta_2^w \neq \beta_2^b\right) \cup \left(\beta_3^w \neq \beta_3^b\right)$$

We use the Chow test, with the following F-test statistic and limiting distribution

$$F = \frac{\left[SSR - \left(SSR_1 + SSR_2\right)\right] / (k+1)}{\left(SSR_1 + SSR_2\right) / \left(n - 2\,(k+1)\right)} \sim F_{k+1,n-2(k+1)}, \tag{16}$$

where SSR is from the full sample in column (2), SSR_1 and SSR_2 are from the two sub-samples, columns (4) and (5). The observed test statistic is then given by

$$F = \frac{\left[SSR^{(2)} - \left(SSR^{(4)} + SSR^{(5)}\right)\right] / 4}{\left[SSR^{(4)} + SSR^{(5)}\right] / (722 - 8)} = \frac{\left[107986 - (86164 + 12368)\right] / 4}{(86164 + 12368) / 714} = 17.13$$

which is greater than the critical value of the $F_{4,714}$ distribution, $F_{0.999,4,714} = 4.67$, and so the null hypothesis is rejected. We conclude that the coefficients for the black individuals and non-black individuals are different and so race has an effect on how own and parental education influences IQ.

(d) We want to know whether the impact of own and parents education on IQ test score varies by race. Conduct such a test of hypothesis and report your findings. Explain why this test is different from the test in (c).

Solution: Strictly the Chow test in (c) tests equality of all coefficients, including the intercept. Since we are only interested in the three slope coefficients of education, then it may be more appropriate to test the hypothesis of race difference using the following specification

$$\texttt{IQ} = \beta_0 + \beta_1 \texttt{feduc}_i + \beta_2 \texttt{meduc}_i + \beta_3 \texttt{educ}_i + \gamma_1 \texttt{feducb}_i + \gamma_2 \texttt{meducb}_i + \gamma_3 \texttt{educb}_i + u_i,$$

where the additional variables are the interactions of the educational variables with the race dummy, namely $\texttt{feducb}_i = \texttt{feduc}_i \times \texttt{black}_i$, $\texttt{meducb}_i = \texttt{meduc}_i \times \texttt{black}_i$ and $\texttt{educb}_i = \texttt{educ}_i \times \texttt{black}_i$. Hence, the coefficients γ_1, γ_2 and γ_3 show the additional partial effect of father's education, mother's education and own education for black individuals as compared to non-black individuals. Therefore, in order to test whether race effects IQ returns to education we would like to test the following hypothesis

$$H_0\colon (\gamma_1 = 0) \cap (\gamma_2 = 0) \cap (\gamma_3 = 0)$$
$$H_1\colon (\gamma_1 \neq 0) \cup (\gamma_2 \neq 0) \cup (\gamma_3 \neq 0)$$

We then estimate the regression with the interactive dummies using `STATA` commands

```
>> gen educb = educ*black
>> gen feducb = feduc*black
>> gen meducb = meduc*black
```

```
>> reg IQ feduc meduc educ feducb meducb educb
```

The output is presented in column (6) of Table 2. Conduct the F-test where we treat the augmented regression above as the unrestricted regression and regression in column (2) as the restricted regression. Then use the following F-test statistic

$$F = \frac{\left(SSR^{(2)} - SSR^{(6)}\right)/q}{SSR^{(6)}/\left(n-k-1\right)} = \frac{\left(107986 - 98756\right)/3}{98756/715} = 22.28.$$

Under the null, $F \sim F_{3,715}$ with $q = 3$ restrictions and $n - k - 1 = 715$. The critical value at $\alpha = 0.001$ is given by $F_{0.999,3,715} = 5.48$. The test statistic is greater than the critical value, so we reject the null and once again conclude that race has an effect on the way education affects IQ test score. Instead of doing the F-test by hand, we can also use the following `STATA` command to conduct a test for joint significance of the variables `feducb`, `meducb`, and `educb`, which of course yield the identical result.

```
>> reg IQ feduc meduc educ feducb meducb educb
>> test feducb meducb educb
```

S3.3 Don't skip conditioning training, it's essential for your overall metrics. Revisit the problem from Supervision 2 and consider the following regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i,\ i = 1, \ldots, n. \tag{17}$$

Suppose the error terms u_i are iid for $i = 1, \ldots, n$, such that $\mathbb{E}\left[u_i|\mathbf{X}\right] = 0$ and $\mathrm{Var}\left(u_i|\mathbf{X}\right) = \sigma^2$, where $\mathbf{X} = (X_1, \ldots, X_n)^\intercal$ is the vector of observations for the explanatory variable, which are assumed to be random variables. Let $\hat{\beta}_1$ denote the OLS estimator of β_1. Consider another estimator $\tilde{\beta}_1$ of β_1, constructed in the following way, assuming $X_3 + X_1 - 2X_2 \neq 0$:

$$\tilde{\beta}_1 = \frac{Y_3 + Y_1 - 2Y_2}{X_3 + X_1 - 2X_2}. \tag{18}$$

(a) Is $\tilde{\beta}_1$ a consistent estimator of β_1?

Solution: First, observe that $\tilde{\beta}_1$ does not change as more data are added, and so it remains a random variable as $n \to \infty$, that is it does not converge to β_1. Also, from last supervision we know that

$$\mathrm{Var}\left(\tilde{\beta}\right) = 6\sigma^2 \mathbb{E}\left[\frac{1}{\left(X_3 + X_1 - 2X_2\right)^2}\right] \tag{19}$$

which does not converge to zero as $n \to \infty$, a condition necessary for consistency.

(b) Show that the OLS estimator $\hat{\beta}_1$ given by

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) Y_i}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2} \tag{20}$$

is unconditionally unbiased for β_1. Carefully explain which Gauss-Markov assumptions are important for showing unbiasedness. First, show that the OLS estimator can be expressed as

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) u_i}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2}. \tag{21}$$

Use that to compute conditional expectation $\mathbb{E}[\hat{\beta}_1|\mathbf{X}]$. Finally apply the law of iterated expectations to find $\mathbb{E}[\hat{\beta}_1]$

Solution: Consider (20) and substitute the true model for Y_i

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) Y_i}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2} \\
&= \frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) \left(\beta_0 + \beta_1 X_i + u_i\right)}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2} \\
&= \beta_0 \left[\frac{\sum_{i=1}^n \left(X_i - \bar{X}\right)}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2}\right] + \beta_1 \left[\frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) X_i}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2}\right] + \frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) u_i}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2}.
\end{aligned}$$

The first term is zero since $\sum_{i=1}^n \left(X_i - \bar{X}\right) = n\bar{X} - n\bar{X} = 0$, the second term is β_1 since the summations cancel out as we know that

$$\sum_{i=1}^n \left(X_i - \bar{X}\right) X_i = \sum_{i=1}^n \left(X_i - \bar{X}\right) \left(X_i - \bar{X}\right).$$

The OLS estimator is then expressed in the form that must be familiar to you by now

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) u_i}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2}. \tag{22}$$

Now, in order to show unbiasedness, we could assume that $\mathbf{X}$ are fixed variables and take the unconditional expectation of the right-hand and be done. However, this would be too easy and no fun at all. Instead, recall that we assumed X_i to be random variables. In this case, we have to first show that $\hat{\beta}_1$ is conditionally unbiased, and then use the law of iterated expectations to show that it is also unconditionally unbiased. To show this, let's first derive $\mathbb{E}[\hat{\beta}_1|\mathbf{X}] = \mathbb{E}[\hat{\beta}_1|X_1, \ldots, X_n]$:

$$\begin{aligned}
\mathbb{E}\left[\hat{\beta}_1|\mathbf{X}\right] &= \beta_1 + \mathbb{E}\left[\frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) u_i}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2}|\mathbf{X}\right] \\
&= \beta_1 + \frac{1}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2} \mathbb{E}\left[\sum_{i=1}^n \left(X_i - \bar{X}\right) u_i|\mathbf{X}\right] \\
&= \beta_1 + \frac{1}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2} \left(\mathbb{E}\left[\sum_{i=1}^n X_i u_i|\mathbf{X}\right] - \mathbb{E}\left[\sum_{i=1}^n \bar{X} u_i|\mathbf{X}\right]\right) \\
&= \beta_1 + \frac{1}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2} \left(\sum_{i=1}^n \mathbb{E}\left[X_i u_i|\mathbf{X}\right] - \sum_{i=1}^n \mathbb{E}\left[\bar{X} u_i|\mathbf{X}\right]\right) \\
&= \beta_1 + \frac{1}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2} \left(\sum_{i=1}^n X_i \mathbb{E}\left[u_i|\mathbf{X}\right] - \sum_{i=1}^n \bar{X} \mathbb{E}\left[u_i|\mathbf{X}\right]\right) \\
&= \beta_1.
\end{aligned}$$

In this derivation, we needed the Gauss-Markov assumption (A3) of exogeneity to hold, i.e. $\mathbb{E}\left[u_i|X_i\right] = 0$ for all $i = 1, \ldots, n$. Also, since X_i are iid, $\mathbb{E}\left[u_i|\mathbf{X}\right] =$

$\mathbb{E}[u_i|X_i] = 0$, as none of the other X_j s.t. $j \neq i$ influence u_i. Using the law of iterated expectations we can then show that $\mathbb{E}[u_i] = \mathbb{E}[\mathbb{E}[u_i|X_i]] = 0$ and hence that

$$\mathbb{E}[u_i X_i] = \mathbb{E}[\mathbb{E}[u_i X_i|X_i]] = \mathbb{E}[X_i \mathbb{E}[u_i|X_i]] = 0. \tag{23}$$

This holds for all $i = 1, \ldots, n$. As a consequence, all the terms that involve the expectation in the second component of the derivation above become zero. The result for $\mathbb{E}[\hat{\beta}_1|\mathbf{X}] = \beta_1$ follows immediately. Finally, apply the law of iterated expectations one more time to show unconditional unbiasedness:

$$\mathbb{E}\left[\hat{\beta}_1\right] = \mathbb{E}\left[\mathbb{E}\left[\hat{\beta}_1|\mathbf{X}\right]\right] = \mathbb{E}[\beta_1] = \beta_1. \tag{24}$$

(c) Show that the OLS estimator $\hat{\beta}_1$ is consistent for β_1. Carefully explain which Gauss-Markov assumptions are important for showing consistency. [Hint: consider the same expression for $\hat{\beta}_1$ as in (b) and use the law of large numbers for the numerator and denominator separately.]

Solution: To show consistency ($\hat{\beta}_1 \to \beta_1$ as $n \to \infty$), first divide the top and the bottom of (22) by n:

$$\hat{\beta}_1 = \beta_1 + \frac{\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)u_i}{\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2} = \beta_1 + \frac{S_{Xu}}{S_{XX}}. \tag{25}$$

Let's examine the numerator and the denominator of the second term in (25). The numerator S_{Xu} is the sample covariance between X_i and u_i. The denominator S_{XX} is the sample variance of X. To show consistency, we want to argue that the term S_{Xu}/S_{XX} converges to zero as n tends to infinity. First, use $\mathbb{E}[u_i] = 0$ and $\mathbb{E}[X_i u_i] = 0$ from (23) to show that the population covariance of X_i and u_i is zero:

$$\mathrm{Cov}(X_i, u_i) = \mathbb{E}[X_i u_i] - \mathbb{E}[X_i]\,\mathbb{E}[u_i] = 0. \tag{26}$$

Now, in order to find the probability limit of $\hat{\beta}_1$ as $n \to \infty$, notice that since X_i and u_i are iid random variables, that are independent from each other, $\left(X_i - \bar{X}\right)u_i$ are iid and similarly $\left(X_i - \bar{X}\right)^2$ are iid for all $i = 1, \ldots, n$, provided the Gauss-Markov assumption of no serial correlation is satisfied. Hence, we can apply the law of large numbers separately to the top and bottom of the second term in (22), to show that as $n \to \infty$,

$$S_{Xu} = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)u_i \longrightarrow \mathbb{E}\left[\left(X_i - \bar{X}\right)u_i\right] = \mathrm{Cov}(X_i, u_i) = 0, \tag{27}$$

$$S_{XX} = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \longrightarrow \mathbb{E}\left[\left(X_i - \bar{X}\right)^2\right] = \mathrm{Var}(X_i) < \infty. \tag{28}$$

Here we need to assume that there is some finite variance in the explanatory variable X, say $\mathrm{Var}(X_i) = \sigma_x^2 < \infty$. Combine these results to conclude that as $n \to \infty$.

$$\hat{\beta}_1 = \beta_1 + \frac{\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)u_i}{\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2} \longrightarrow \beta_1 + \frac{\mathrm{Cov}(X_i, u_i)}{\mathrm{Var}(X_i)} = \beta_1.$$

Formally, to claim convergence, we need to use something called the Slutsky's theorem in order to make sure that the limit of a quotient in (22) is the quotient of the limits.

S3.4 Sir Francis Galton brings back the conditional expectation function. Consider the following conditional expectation function

$$Y_i = \mathbb{E}\left[Y_i|X_i\right] + u_i, \quad i = 1, \ldots, n, \tag{29}$$

where Y_i and X_i are random variables, and u_i is random variable with mean zero and constant variance, independent of X_i, such that $\mathbb{E}\left[u_i|X_i\right] = 0$, for all i. The linear regression is a special case of (29), where $\mathbb{E}\left[Y_i|X_i\right] = \beta_0 + \beta_1 X_i$. You can then think of a linear regression as a linear conditional expectation function

$$r(x) = \mathbb{E}\left[Y_i|X_i = x\right] = \beta_0 + \beta_1 x. \tag{30}$$

This says that if $X_i = x$, then the expected value of Y_i conditional on X_i is $\beta_0 + \beta_1 x$. Note that we are again relaxing assumption (A2) of fixed regressors.

(a) Use the law of iterated expectations to show that $\mathbb{E}\left[u_i\right] = 0$, $\mathbb{E}\left[X_i u_i\right] = 0$ and $\mathrm{Cov}\left(X_i, u_i\right) = 0$.

Solution: Use the LIE and the assumption about $\mathbb{E}\left[u_i|X_i\right] = 0$ in the following:

$$\begin{aligned}
\mathbb{E}\left[u_i\right] &= \mathbb{E}\left[\mathbb{E}\left[u_i|X_i\right]\right] = \mathbb{E}\left[0\right] = 0 \\
\mathbb{E}\left[X_i u_i\right] &= \mathbb{E}\left[\mathbb{E}\left[X_i u_i|X_i\right]\right] = \mathbb{E}\left[X_i \mathbb{E}\left[u_i|X_i\right]\right] = \mathbb{E}\left[X_i \times 0\right] = 0 \\
\mathrm{Cov}\left(X_i, u_i\right) &= \mathbb{E}\left[X_i, u_i\right] - \mathbb{E}\left[X_i\right]\mathbb{E}\left[u_i\right] = 0
\end{aligned}$$

(b) Show that the unconditional variance of the dependent variable can be written as

$$\mathrm{Var}\left(Y_i\right) = \mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right]\right) + \mathbb{E}\left[u_i^2\right]. \tag{31}$$

Interpret this expression with the reference to the explained and the unexplained components of Y_i. [Hint: denote $Y_i^* = \mathbb{E}\left[Y_i|X_i\right]$ and observe that Y_i^* is a random variable. Then recognize that $Y_i = Y_i^* + u_i$, where Y_i^* is the explained part of Y_i and u_i is the unexplained part. So compute $\mathrm{Var}\left(Y_i^* + u_i\right)$.]

Solution: Denote the conditional expectation of Y_i given X_i by $Y_i^* = \mathbb{E}\left[Y_i|X_i\right]$. Expression (29) then implies $Y_i = Y_i^* + u_i$. This means that Y_i can be decomposed into two component. The first is the explained part Y_i^*, the component that can be conditionally explained by X_i. The second is the unexplained part u_i, the component that cannot be conditionally explained by X_i. Our goal in this question is to derive the variance of a sum of two random variables, namely $\mathrm{Var}\left(Y_i\right) = \mathrm{Var}\left(Y_i^* + u_i\right)$ and show that $\mathrm{Var}\left(Y_i\right) = \mathrm{Var}\left(Y_i^*\right) + \mathrm{Var}\left(u_i\right)$. I will derive this explicitly, without making the substitution for Y_i^*.

$$\begin{aligned}
\mathrm{Var}\left(Y_i\right) &= \mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right] + u_i\right) \\
[\![\text{formula for } \mathrm{Var}\left(Y_i^* + u_i\right)]\!] &= \mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right]\right) + \mathrm{Var}\left(u_i\right) + 2\mathrm{Cov}\left(\mathbb{E}\left[Y_i|X_i\right], u_i\right) \\
[\![u_i \text{ independent of } X_i]\!] &= \mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right]\right) + \mathrm{Var}\left(u_i\right) \\
[\![\text{formula for } \mathrm{Var}\left(u_i\right)]\!] &= \mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right]\right) + \mathbb{E}\left[u_i^2\right] - \left(\mathbb{E}\left[u_i\right]\right)^2 \\
[\![\mathbb{E}\left[u_i\right] = 0]\!] &= \mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right]\right) + \mathbb{E}\left[u_i^2\right]
\end{aligned}$$

This expression says that the variance of the dependent variable Y_i can be decomposed into two components. The first is the variance of the conditional expectation function $Y_i^* = \mathbb{E}\left[Y_i|X_i\right]$, which is the part of Y_i that can be explained by the regressor X_i.

The second is the variance of the error term u_i, which is the part of Y_i that cannot be explained by X_i. Quite intuitively, since the explained and the unexplained parts are independent and hence uncorrelated, the variance of the variable that you are trying to explain is the sum of the variance of the part of Y_i that you can explain conditionally, and the variance of the part of Y_i that you cannot explain. We can write this explicitly as

$$\mathrm{Var}\left(Y_i\right) = \mathrm{Var}\left(Y_i^* + u_i\right) = \mathrm{Var}\left(Y_i^*\right) + \mathrm{Var}\left(u_i\right) = \mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right]\right) + \mathbb{E}\left[u_i^2\right]. \tag{32}$$

Recall the general formula for the variance of the sum of two random variables

$$\mathrm{Var}\left(Y_i^* + u_i\right) = \mathrm{Var}\left(Y_i^*\right) + \mathrm{Var}\left(u_i\right) + 2\mathrm{Cov}\left(Y_i^*, u_i\right).$$

Hence, it must be the case that $\mathrm{Cov}\left(Y_i^*, u_i\right) = 0$. Moreover, we have shown that Y_i^* and u_i are not only uncorrelated but also independent, since

$$\mathbb{E}\left[Y_i^* u_i\right] = \mathbb{E}\left[\mathbb{E}\left[Y_i^* u_i|X_i\right]\right] = \mathbb{E}\left[\mathbb{E}\left[\mathbb{E}\left[Y_i|X_i\right] u_i|X_i\right]\right] = \mathbb{E}\left[\mathbb{E}\left[Y_i|X_i\right] \mathbb{E}\left[u_i|X_i\right]\right] = 0$$

(c) Show that $\mathbb{E}\left[u_i^2|X_i\right] = \mathrm{Var}\left(Y_i|X_i\right)$. [Hint: substitute in for $u_i = Y_i - \mathbb{E}\left[Y_i|X_i\right]$ and then use the law of iterated expectations where necessary.]

Solution: Proceed very carefully:

$$\begin{aligned}
\mathbb{E}\left[u_i^2|X_i\right] &= \mathbb{E}\left[\left(Y_i - \mathbb{E}\left[Y_i|X_i\right]\right)^2|X_i\right] \\
[\![\text{expand the square}]\!] &= \mathbb{E}\left[Y_i^2 - 2\mathbb{E}\left[Y_i|X_i\right] Y_i + \left(\mathbb{E}\left[Y_i|X_i\right]\right)^2|X_i\right] \\
[\![\text{expectation is linear}]\!] &= \mathbb{E}\left[Y_i^2|X_i\right] - \mathbb{E}\left[2\mathbb{E}\left[Y_i|X_i\right] Y_i|X_i\right] + \mathbb{E}\left[\left(\mathbb{E}\left[Y_i|X_i\right]\right)^2|X_i\right] \\
[\![\text{LIE}]\!] &= \mathbb{E}\left[Y_i^2|X_i\right] - 2\mathbb{E}\left[Y_i|X_i\right] \mathbb{E}\left[Y_i|X_i\right] + \left(\mathbb{E}\left[Y_i|X_i\right]\right)^2 \\
&= \mathbb{E}\left[Y_i^2|X_i\right] - 2\left(\mathbb{E}\left[Y_i|X_i\right]\right)^2 + \left(\mathbb{E}\left[Y_i|X_i\right]\right)^2 \\
&= \mathbb{E}\left[Y_i^2|X_i\right] - \left(\mathbb{E}\left[Y_i|X_i\right]\right)^2 \\
&= \mathrm{Var}\left(Y_i|X_i\right)
\end{aligned}$$

This simply says that variance of Y_i conditional on the explanatory variable X_i is equal to conditional variance of the error term u_i.

(d) Conclude that $\mathrm{Var}\left(Y_i\right) = \mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right]\right) + \mathbb{E}\left[\mathrm{Var}\left(Y_i|X_i\right)\right]$. Interpret this equation.

Solution: The last step is to apply the law of iterated expectations on $\mathbb{E}\left[u_i^2\right]$:

$$\mathbb{E}\left[u_i^2\right] = \mathbb{E}\left[\mathbb{E}\left[u_i^2|X_i\right]\right] = \mathbb{E}\left[\mathrm{Var}\left(Y_i|X_i\right)\right]. \tag{33}$$

The overall result follows immediately:

$$\mathrm{Var}\left(Y_i\right) = \mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right]\right) + \mathbb{E}\left[\mathrm{Var}\left(Y_i|X_i\right)\right]. \tag{34}$$

As before, the first term in this equation, $\mathrm{Var}\left(\mathbb{E}\left[Y_i|X_i\right]\right)$, is the variation of the component of Y_i that can be explained by X_i, i.e. the variance of the conditional expectation function. The second term on the right-hand side of equation (34) can be interpreted as follows. Having conditioned Y_i on X_i, we have effectively taken out the part of Y_i that can be explained by the model. The remaining variability in Y_i has to come from its unexplained component, which is u_i. Hence, the term $\mathbb{E}\left[\mathrm{Var}\left(Y_i|X_i\right)\right]$ is

just the average (expected) variability of the unexplained part of Y_i. Equation (34) is the same decomposition as before, recall part (b). The only difference is that now it is represented in different notation with the reference to the conditional expectation function only. We can therefore conclude that total variability in Y_i depends only on the conditional expectation function. Quite neat, don't you find?

(e) Now, suppose that we are dealing with a linear regression model, where

$$Y_i = \mathbb{E}[Y_i|X_i] + u_i = \beta_0 + \beta_1 X_i + u_i. \tag{35}$$

Show that $\beta_0 = \mathbb{E}[Y_i] - \beta_1 \mathbb{E}[X_i]$ and $\beta_1 = \text{Cov}(Y_i, X_i)/\text{Var}(X_i)$.
Solution: Take conditional expectation on both sides of $Y_i = \beta_0 + \beta_1 X_i + u_i$:

$$\mathbb{E}[Y_i|X_i] = \mathbb{E}[\beta_0 + \beta_1 X_i + u_i|X_i] = \mathbb{E}[\beta_0|X_i] + \mathbb{E}[\beta_1 X_i|X_i] + \mathbb{E}[u_i|X_i].$$

Conclude that $\mathbb{E}[Y_i|X_i] = \beta_0 + \beta_1 X_i$. Now, according to the law of iterated expectations $\mathbb{E}[\mathbb{E}[Y_i|X_i]] = \mathbb{E}[Y_i]$. Use this when we take the expectation on both sides of the expression above:

$$\mathbb{E}[Y_i] = \mathbb{E}[\mathbb{E}[Y_i|X_i]] = \mathbb{E}[\beta_0 + \beta_1 X_i] \implies \beta_0 = \mathbb{E}[Y_i] - \beta_1 \mathbb{E}[X_i]. \tag{36}$$

Finally, take covariance of both sides of (35) with X_i:

$$\begin{aligned}\text{Cov}(Y_i, X_i) &= \text{Cov}(\beta_0 + \beta_1 X_i + u_i, X_i) \\ &= \text{Cov}(\beta_0, X_i) + \beta_1 \text{Cov}(X_i, X_i) + \text{Cov}(u_i, X_i).\end{aligned}$$

The first term and last terms are zero, while the second term is $\beta_1 \text{Var}(X_i)$, hence $\text{Cov}(Y_i, X_i) = \beta_1 \text{Var}(X_i)$. Rearrange this for β_1 and conclude that

$$\beta_1 = \frac{\text{Cov}(Y_i, X_i)}{\text{Var}(X_i)}. \tag{37}$$

S3.5 Unbiased estimators do not discriminate against heteroskedastic errors.
A researcher is trying to estimate mean household income, μ, for a sample of 100 households. He knows that the households can be divided into two groups of equal size: those with "safe" jobs the and those with "risky" jobs:

$$\begin{aligned}&\text{safe jobs:} && Y_i = \mu + \varepsilon_i, && \varepsilon_i \sim \text{iid}(0,1), && i = 1, \ldots, 50 \\ &\text{risky jobs:} && Y_i = \mu + v_i, && v_i \sim \text{iid}(0,4), && i = 51, \ldots, 100\end{aligned}$$

The error terms are assumed to be independent, $\text{Cov}(\varepsilon_i, v_j) = 0$ for all i and j.

(a) Show that the sample mean $\bar{Y}$ is an unbiased estimator of the population mean μ.
Solution: Compute the sample mean $\bar{Y}$ and then its expectation. The sample of size $n = 100$ is divided into two sub-samples of equal sizes, $n_s = n_r = 50$. The first 50 observations correspond to safe jobs, $Y_i = \mu + \varepsilon_i$ for $i = 1, \ldots, 50$. The last 50 observations correspond to risky jobs, $Y_i = \mu + v_i$ for $i = 51, \ldots, 100$. The overall sample mean is then given by

$$\bar{Y} = \frac{1}{100}\sum_{i=1}^{100} Y_i = \frac{1}{100}\left[\sum_{i=1}^{50}(\mu + \varepsilon_i) + \sum_{i=51}^{100}(\mu + v_i)\right] = \mu + \frac{1}{100}\left(\sum_{i=1}^{50}\varepsilon_i + \sum_{i=51}^{100} v_i\right).$$

Since $\mathbb{E}[\varepsilon_i] = \mathbb{E}[v_i] = 0$, this is unbiased for μ:

$$\mathbb{E}\left[\bar{Y}\right] = \mathbb{E}[\mu] + \frac{1}{100}\left(\sum_{i=1}^{50}\mathbb{E}[\varepsilon_i] + \sum_{i=51}^{100}\mathbb{E}[v_i]\right) = \mu. \tag{38}$$

We can visualize this type of data using Figure 2. For simplicity, pick $\mu = 10$ and normal error terms, $\varepsilon_i \sim \text{iid}\, N(0,1)$ and $v_i \sim \text{iid}\, N(0,4)$. Note that the first 50 observations of Y_i (empty dots) have lower variance than the last 50 observations (solid dots). However, for both subsets the mean is exactly the same and is represented by the solid line, $\mu = 10$. This is an example of the dataset generated using heteroskedastic errors, that is errors whose variance is not constant for all observations.

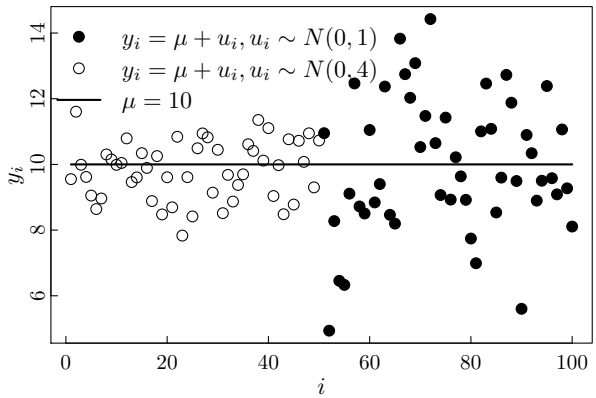

Figure 2: Income data y_i generated with heteroskedastic errors.

(b) Show that the sample mean is not an efficient estimator of μ by deriving an unbiased estimator with smaller variance.

Solution: Find the variance of $\bar{Y}$, where we need to assume that ε_i are independent of v_j for all i and j:

$$\begin{aligned}
\text{Var}\left(\bar{Y}\right) &= \text{Var}\left(\mu + \frac{1}{100}\left(\sum_{i=1}^{50}\varepsilon_i + \sum_{i=51}^{100} v_i\right)\right) \\
[\![\text{assuming Cov}(\varepsilon_i, v_j) = 0, \text{ all } i,j]\!] &= \frac{1}{100^2}\left[\text{Var}\left(\sum_{i=1}^{50}\varepsilon_i\right) + \text{Var}\left(\sum_{i=51}^{100} v_i\right)\right] \\
[\![\text{Cov}(\varepsilon_i, \varepsilon_j) = \text{Cov}(v_i, v_j) = 0, i \neq j]\!] &= \frac{1}{100^2}\left[\sum_{i=1}^{50}\text{Var}(\varepsilon_i) + \sum_{i=51}^{100}\text{Var}(v_i)\right] \\
&= \frac{1}{100^2}\sum_{i=1}^{50}\Big[\text{Var}(\varepsilon_i) + \text{Var}(v_i)\Big] \\
&= \frac{1}{100^2} \times 50 \times (1+4) \\
&= \frac{1}{40}.
\end{aligned}$$

Observe that the sample mean for the safe jobs $\bar{Y}_s = \frac{1}{50}\sum_{i=1}^{50}(\mu+\varepsilon_i)$ is still an unbiased estimator of μ, but has variance smaller that that of the overall mean $\bar{Y}$:

$$\mathrm{Var}\left(\bar{Y}_s\right) = \mathrm{Var}\left(\mu + \frac{1}{50}\sum_{i=1}^{50}\varepsilon_i\right) = \frac{1}{50^2}\sum_{i=1}^{50}\mathrm{Var}\left(\varepsilon_i\right) = \frac{1}{50} < \frac{1}{40} = \mathrm{Var}\left(\bar{Y}\right).$$

Note also that the sample mean for the risky jobs, $\bar{Y}_r$ will be unbiased, but its variance is higher than that of the overall sample mean, $\mathrm{Var}\left(\bar{Y}_r\right) = \frac{1}{12}$.

(c) Discuss the implications of these results for the use of the Ordinary Least Squares estimators in linear regression analysis.

Solution: In a linear regression between two variables y_i and x_i,

$$y_i = \beta_0 + \beta_1 x_i + u_i, \quad i = 1, \ldots, n, \tag{39}$$

the Gauss-Markov assumptions say that $u_i \sim \text{iid}\left(0, \sigma^2\right)$. This means that the errors are iid with zero mean and constant variance (homoskedastic). If this is the case, the theorem says that the OLS estimators of β_0 and β_1 are unbiased and efficient, that is have lowest variance among all linear unbiased estimators. The assumption of homoskedasticity is necessary for efficiency (but not for unbiasedness). If the homoskedasticity assumption is violated, and the error terms do not have constant variance, we say they are heteroskedastic. In this case, the OLS estimators will no longer be efficient (but still unbiased). Then, it is possible to find a linear unbiased estimator with a smaller variance. This is exactly the setup in this question, where the error term does not have constant variance. To see this more explicitly, instead of the original notation with ε_i and v_i, we could be express the error terms as random variables u_i, $i = 1, \ldots, n$, which are iid with $\mathbb{E}\left[u_i\right] = 0$ for all $i = 1, \ldots, n$ and

$$\mathrm{Var}\left(u_i\right) = \begin{cases} 1 & \text{for } i = 1, \ldots, 50 \\ 4 & \text{for } i = 51, \ldots, 100 \end{cases}$$

We will consider ways of dealing with heteroscedasticity later in the course.

S3.7 Dummy Wars. Episode VI: Return of the education.

Consider the following model for the determination of individual earnings:

$$\texttt{lwage}_i = \beta_0 + \beta_1\texttt{male}_i + \beta_2\texttt{age}_i + \beta_3\texttt{IQ}_i + \beta_4\texttt{educ}_i + \varepsilon_i, \quad i = 1, \ldots, n \tag{40}$$

where ε_i are iid error terms with mean zero and constant variance, and the explanatory variables are defined as follows:

`lwage`	logarithm of hourly earnings for individual i
`male`	dummy variable which takes value 1 for males and 0 otherwise
`age`	age of individual i (measured in years)
`IQ`	measure of the ability of individual i
`educ`	the schooling of individual i (measured in years)

(a) Suppose you are interested in the hypothesis that the effect of schooling on the logarithm of hourly earnings is nonlinear. Describe a simple procedure that you could

follow to test this hypothesis, assuming that $\texttt{educ}_i$ can be treated as continuous variable. Does your test procedure change depending on whether the error terms ε_i are normally distributed or not?

Solution: Regression (40) suggests that the effect of education on log earnings is linear. If we assume that $\texttt{educ}_i$ is approximately continuous, then the interpretation of the corresponding slope coefficient is the proportional change in wage with respect to a small change in education:

$$\beta_4 = \left.\frac{\partial \texttt{lwage}_i}{\partial \texttt{educ}_i}\right|_{\texttt{male}_i,\texttt{age}_i,\texttt{IQ}_i} \tag{41}$$

A simple procedure to test whether the effect of education on log wage is non-linear could be based on estimating a regression with the addition of a squared education as an explanatory variable. With this specification we will be able to check whether returns to education are linear, concave or convex.

$$\texttt{lwage}_i = \beta_0 + \beta_1\texttt{male}_i + \beta_2\texttt{age}_i + \beta_3\texttt{IQ}_i + \beta_4\texttt{educ}_i + \beta_5\texttt{educ}_i^2 + \varepsilon_i. \tag{42}$$

In this regression, the partial effect of education depends on the value of education:

$$\left.\frac{\partial \texttt{lwage}_i}{\partial \texttt{educ}_i}\right|_{\texttt{male}_i,\texttt{age}_i,\texttt{IQ}_i} = \beta_4 + \beta_5\texttt{educ}_i. \tag{43}$$

If $\beta_5 > 0$ the the effect is increasing in education: the additional effect on wage will be increasing linearly with education. Intuitively, this would imply that the expected jump in wage for someone who has an undergraduate degree and decides to do a masters degree will be higher than the expected jump in wage for someone who has a high-school diploma and decided to do an undergraduate degree (provided of course that these individuals have identical gender, age and IQ). If $\beta_5 < 0$, the effect is opposite. Otherwise, if $\beta_5 < 0$ then wage is a concave function of education or marginal returns to education are diminishing.

In order to test whether this non-linear effect is relevant, we would like to test significance of $\texttt{educ}_i^2$. Formally, test $H_0\colon \beta_5 = 0$ against $H_1\colon \beta_5 \neq 0$ using a t-test statistic or an F-test, where regression (42) is the unrestricted regression and (40) is the restricted regression. The different between the cases when the error terms ε_i are iid normal or iid non-normal is in the limiting distribution of the test statistic under the null hypothesis. In the first case, which works for any sample size, the limiting distribution is exactly t_{n-k-1} with $k = 5$ explanatory variables in (42). In the second case, provided that the sample is large enough for the CLT to apply, the limiting distribution is approximately (asymptotically) $N(0,1)$. If the sample size is very large, then the two tests are computationally equivalent, since $t_n \to N(0,1)$ as $n \to \infty$. Therefore, in large samples a rather strong assumption of normally distribution errors is not required. The difference is summarized in Table 3 for a general slope coefficient β, its OLS estimator $\hat{\beta}$ and standard error $S_{\hat{\beta}}$.

Table 3: Inference on β using OLS $\hat{\beta}$ for normal and non-normal errors.

Errors distribution	**Sample size**	**Null hypothesis**	**Test statistic**
$\varepsilon_i \sim \text{iid}\, N\left(0, \sigma^2\right)$	any n	$H_0\colon \beta = 0 \implies$	$T = \dfrac{\hat{\beta}}{S_{\hat{\beta}}} \sim t_{n-k-1}$
$\varepsilon_i \sim \text{iid}\, \left(0, \sigma^2\right)$	$n > 30$, CLT	$H_0\colon \beta = 0 \implies$	$T = \dfrac{\hat{\beta}}{S_{\hat{\beta}}} \overset{a}{\sim} N\left(0, 1\right)$

(b) How would you test the hypothesis of non-linear effects of education, given that in reality, $\texttt{educ}_i$ is not continuous and can only take values that are natural numbers.

Solution: A more realistic view on $\texttt{educ}_i$ is that it is not a continuous variable and can only admit natural numbers, $\texttt{educ}_i = 0, 1, 2, \ldots a_{max}$, bounded by some upper limit on the possible number of years spent in education, for simplicity say $a_{max} = 20$. Therefore, the partial derivative interpretation of β_4 in (43) is not strictly correct. A more rigorous interpretation would be

$$\beta_4 = \mathbb{E}\left[\texttt{lwage}_i \,|\, \texttt{male}_i, \texttt{age}_i, \texttt{IQ}_i, \texttt{educ}_i = a + 1\right] - \mathbb{E}\left[\texttt{lwage}_i \,|\, \texttt{male}_i, \texttt{age}_i, \texttt{IQ}_i, \texttt{educ}_i = a\right]$$

for any value of $a = 0, 1, 2, \ldots$. This is the difference in expected log wage for an additional year of education for an individual with a given gender, age and IQ. We have already considered an identical problems S1.7 and S2.7. Since β_4 shows an increase in wage that results from a one year increase in education for any current level of attainment, we cannot distinguish the effect of moving from high school to university from the effect of moving from university to an MBA. Hence, we can create 20 dummy variables corresponding to the number of years spent in education, $\texttt{educ_1}_i, \ldots \texttt{educ_20}_i$, that each take the value of 1 if $\texttt{educ}_i$ is equal to that number of years of education and zero otherwise, for example

$$\texttt{educ_10}_i = \begin{cases} 1 & \text{if } \texttt{educ}_i = 10 \\ 0 & \text{otherwise} \end{cases}$$

We then estimate the following regression

$$\texttt{lwage}_i = \beta_0 + \beta_1 \texttt{male}_i + \beta_2 \texttt{age}_i + \beta_3 \texttt{IQ}_i + \sum_{j=1}^{20} \delta_j \texttt{educ_j}_i + \varepsilon_i. \tag{44}$$

Note that the default category, $\texttt{educ}_i = 0$ is not included in the regression to prevent multicollinearity. The δ_j coefficient, $j = 1, \ldots, 20$, in this regression is then interpreted as the additional effect on log wages of j years of education compared to no education at all, that is when $\texttt{educ}_i = 0$, for an individual of a given gender, age and IQ:

$$\delta_j = \mathbb{E}\left[\texttt{lwage}_i \,|\, \texttt{male}_i, \texttt{age}_i, \texttt{IQ}_i, \texttt{educ}_i = j\right] - \mathbb{E}\left[\texttt{lwage}_i \,|\, \texttt{male}_i, \texttt{age}_i, \texttt{IQ}_i, \texttt{educ}_i = 0\right].$$

Hence, the additional log wage from an additional year of education for someone with the current attainment level of, say, 10 years is given by $\delta_{11} - \delta_{10}$. Note that β_4 is the average of all δ_j. In regression (44) we would initially like to test whether δ_j are significantly different from zero, that is $H_0\colon \delta_j = 0$, $j = 1, \ldots, 20$. To test whether

the effect of education is non-linear, we would then want to test a series of hypothesis of the form $H_0\colon \delta_j = \delta_k$ against $H_1\colon \delta_j \neq \delta_k$, for $j \neq k$. We could also test the joint hypothesis that $H_1\colon \delta_1 = \delta_2 = \cdots = \delta_{20}$, to test whether the effects are exactly linear for all levels of educational attainment.

Alternatively, we could run a regression with dummy variables defined for the levels of education, rather than individual years. For instance we could consider $\texttt{midsch}_i$ that takes the value of 1 for GCSE graduates, $\texttt{highsch}_i$ equal to 1 for A-levels graduates, $\texttt{ug}_i$ and $\texttt{pg}_i$ equal to 1 for the holders of undergraduate and postgraduate degrees, respectively.

(c) Suppose in addition that the observations in your dataset are sorted in ascending order of education, so that the first observation refers to the least educated individual and so on. Explain how you could exploit this ordering to generate an additional test of the hypothesis that the effect of schooling is nonlinear.

Solution: A scatter plot for hypothetical data on log wages and education is plotted in an ascending order of education in Figure 3. In addition to the data points $(\texttt{lwage}_i, \texttt{educ}_i)$, the Figure plots the fitted lines (predicted values) from the original linear regression (40), $\widehat{\texttt{lwage}}_L$ in red, from regression (42) with the squared term in education, $\widehat{\texttt{lwage}}_S$, in green, and from regression (44) with the dummies for education, $\widehat{\texttt{lwage}}_D$, in black. These are the conditional expectations of log wage for the given level of education of j years, $\mathbb{E}\left[\texttt{lwage}_i \mid \texttt{educ}_i = j\right]$.

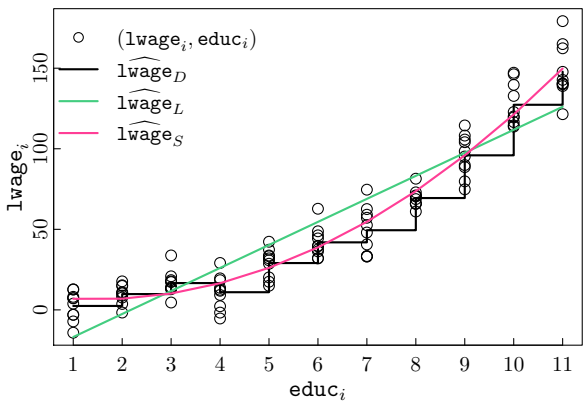

Figure 3: Scatter plot of $(\texttt{lwage}_i, \texttt{educ}_i)$ and predicted $\texttt{lwage}_i$ from (40), $\widehat{\texttt{lwage}}_L$, regression (42), $\widehat{\texttt{lwage}}_S$ and regression (44), $\widehat{\texttt{lwage}}_D$.

The dummy regression (black) provides a much better fit for the data points than the other two models. The red line (linear regression) simply averages out the effects of education across all levels of education. The green line (regression with squared education) provides a slightly better fit, suggesting that returns to education are increasing (fitted line is curving upwards). Moreover, observe that the slope of the fitted line for the original regression is just β_4, while the coefficients δ_j from the regression with dummies vary for every value of $\texttt{educ}_i$. For instance, the jump $\texttt{educ}_i = 1$ to $\texttt{educ}_i = 3$ produces virtually no change in log wage, while a jump from $\texttt{educ}_i = 9$ to $\texttt{educ}_i = 10$ is by far the largest. This graph clearly illustrates the idea that the effects in education are non-linear. Note that you can use this graph to define the

dummy variable across education levels that produce equal effects. For example, `educ_1`, `educ_2`, `educ_3` and `educ_4` could be grouped into one category, since they appear to have the same effect on wages.

(d) It is suspected that your equation is misspecified because it does not allow for the possibility that earnings are also influenced by on-the-job experience. However, you have no data on experience. As a remedy, since children start education at age 5, experience is proxied using the relationship $\texttt{jobex}_i = \texttt{age}_i - \texttt{educ}_i - 5$. What would you expect to find if you added this variable to your original regression?

Solution: It is suggested that $\texttt{jobex}_i = \texttt{age}_i - \texttt{educ}_i - 5$ is added as an extra explanatory variable in the original regression (40). Clearly, $\texttt{jobex}_i$ is a linear function of the explanatory variables that already enter the regression. Hence, adding it to the regression will cause the problem of multicollinearity and the slope coefficients will not be estimable using OLS. If you run a regression with collinear regressors in STATA, the program will stop this immediately and drop one of the variable from the regression. This is because the individual effects of age, education and job experience defined in this way are not distinguishable if all three of variables are included in the regression. Let's show this explicitly by analyzing what happens to the new regression when we add $\texttt{jobex}_i$ as an extra regressor to (40):

$$\begin{aligned}
\texttt{lwage}_i &= \delta_0 + \delta_1\texttt{male}_i + \delta_2\texttt{age}_i + \delta_3\texttt{IQ}_i + \delta_4\texttt{educ}_i + \delta_5\texttt{jobex}_i + \varepsilon_i \\
&= \delta_0 + \delta_1\texttt{male}_i + \delta_2\texttt{age}_i + \delta_3\texttt{IQ}_i + \delta_4\texttt{educ}_i + \delta_5\left(\texttt{age}_i - \texttt{educ}_i - 5\right) + \varepsilon_i \\
&= \left(\delta_0 - 5\delta_5\right) + \delta_1\texttt{male}_i + \left(\delta_2 + \delta_5\right)\texttt{age}_i + \delta_3\texttt{IQ}_i + \left(\delta_4 - \delta_5\right)\texttt{educ}_i + \varepsilon_i \\
&= \beta_0 + \beta_1\texttt{male}_i + \beta_2\texttt{age}_i + \beta_3\texttt{IQ}_i + \beta_4\texttt{educ}_i + \varepsilon_i
\end{aligned}$$

This augmented regression is identical to the original linear regression (40), where only the notation for the coefficients have changed. Overall, we conclude that the addition of $\texttt{jobex}_i$ defined this way has to real explanatory power beyond what `age` and `educ` already explain. Hence, we need to obtain the real measure of on-the-job experience in order to make sure that we don't run in to multicollinearity.